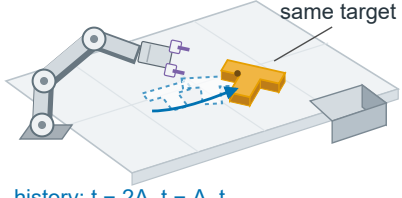
What must be remembered and predicted for interaction?

Conceptual illustration; all positions and trajectories are hypothetical

(a) Remember motion

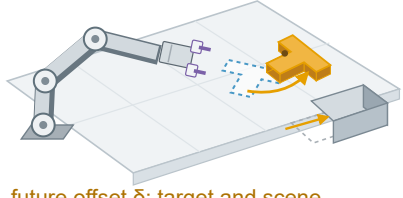


history: t − 2Δ, t − Δ, t

One current pose can hide direction, speed or occlusion.

pose + Δt + visibility

(b) Predict the relevant future



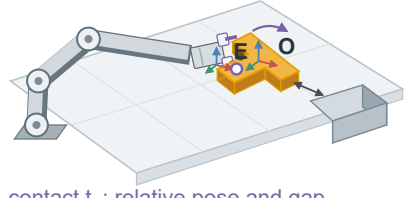
future offset δ: target and scene

target pose

Δp, log ΔR

scene flow / tracks

(c) Resolve contact geometry



contact t_c: relative pose and gap

Target role: align the contact.

Context role: preserve clearance.

O: target; E: end-effector frame